

Lab SHEET 04

(FULL STACK)

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Task 1: Custom Module

```
const fs = require("fs");
const path = require("path");
const os = require("os");
const http = require("http");
const url = require("url");
const EventEmitter = require("events").EventEmitter;
```

// TASK 1: CUSTOM MODULES

```
function sum(a, b) {
  return a + b;}
function evenOdd(num) {
  return num % 2 === 0 ? "Even" : "Odd";}
console.log("\n===== TASK 1 =====");
console.log("Sum:", sum(10, 20));
console.log("7 is:", evenOdd(7));
console.log("12 is:", evenOdd(12));
```

```
===== TASK 1 =====
Sum: 30
7 is: Odd
12 is: Even
PS D:\full stack>
```

// // TASK 2: FILE SYSTEM

```
console.log("\n===== TASK 2 =====");
fs.writeFileSync(
  "student.txt",
  "Name: Priyanshu\nRoll No: 101\n"
);
fs.appendFileSync(
  "student.txt",
  "Course: Full Stack Web Development\n"
);
console.log("File Content:");
console.log(fs.readFileSync("student.txt", "utf8"));
if (fs.existsSync("profile.txt")) {
  fs.unlinkSync("profile.txt");
```

```

}
fs.renameSync("student.txt", "profile.txt");
console.log("student.txt renamed to profile.txt")

```

```

===== TASK 2 =====
File Content:
Name: Priyanshu
Roll No: 101
Course: Full Stack Web Development

student.txt renamed to profile.txt
> PS D:\full stack>

```

// TASK 3: DIRECTORY OPERATIONS

```

console.log("\n===== TASK 3 =====");
if (!fs.existsSync("uploads")) {
  fs.mkdirSync("uploads");
}
fs.writeFileSync("uploads/file1.txt", "");
fs.writeFileSync("uploads/file2.txt", "");
fs.writeFileSync("uploads/file3.txt", "");
console.log("Files created:");
console.log(fs.readdirSync("uploads"));
if (fs.existsSync("uploads/file3.txt")) {
  fs.unlinkSync("uploads/file3.txt");
}
console.log("file3.txt deleted.");

```

```

===== TASK 3 =====
Files created:
[ 'file1.txt', 'file2.txt', 'file3.txt' ]
file3.txt deleted.
> PS D:\full stack>

```

// TASK 4: PATH MODULE

```

console.log("\n===== TASK 4 =====");
const filePath = "/home/user/data/report.pdf";
console.log("Directory:", path.dirname(filePath));
console.log("Base Name:", path.basename(filePath));
console.log("Extension:", path.extname(filePath));
console.log(
  "Absolute Path:",
  path.resolve("./data/report.pdf")
);

```

```
===== TASK 4 =====
Directory: /home/user/data
Base Name: report.pdf
Extension: .pdf
Absolute Path: D:\full stack\data\report.pdf
PS D:\full stack>
```

// TASK 5: OS MODULE

```
console.log("\n===== TASK 5 =====");
console.log("Platform:", os.platform());
console.log("Architecture:", os.arch());
console.log(
  "Total Memory:",
  (os.totalmem() / 1024 / 1024 / 1024).toFixed(2),
  "GB");
console.log(
  "Free Memory:",
  (os.freemem() / 1024 / 1024 / 1024).toFixed(2),
  "GB"
);
console.log("CPU Cores:", os.cpus().length);
console.log(
  "System Uptime:",
  (os.uptime() / 60).toFixed(2),
  "minutes"
);
===== TASK 5 =====
Platform: win32
Architecture: x64
Total Memory: 7.85 GB
Free Memory: 2.41 GB
CPU Cores: 8
System Uptime: 125.63 minutes
```

// TASK 6: EVENTS MODULE

```
console.log("\n===== TASK 6 =====");
const orderEmitter = new EventEmitter();
orderEmitter.on("orderPlaced", (order) => {
  console.log("\n===== ORDER RECEIPT =====");
  console.log("Order ID:", order.id);
  console.log("Item:", order.item);
  console.log("Amount: ₹" + order.amount);
  console.log("=====");
});
```

```

orderEmitter.emit("orderPlaced", {
  id: 101,
  item: "Laptop",
  amount: 55000});
orderEmitter.emit("orderPlaced", {
  id: 102,
  item: "Mouse",
  amount: 800});
orderEmitter.emit("orderPlaced", {
  id: 103,
  item: "Keyboard",
  amount: 1500});

```

```

PS D:\full_stack> node "d:\ful
===== ORDER RECEIPT =====
Order ID: 102
Item: Mouse
Amount: ₹800
=====

===== ORDER RECEIPT =====
Order ID: 103
Item: Keyboard
Amount: ₹1500
=====
> PS D:\full_stack>

```

// TASK 7 & 8: HTTP SERVER + QUERY STRING

```

console.log("\n===== TASK 7 & 8 =====");
const students = [{
  id: 1,
  name: "Rahul",
  course: "B.Tech" },
{ id: 2,
  name: "Priyanshu",
  course: "BCA"},
{ id: 3,
  name: "Aman",
  course: "MCA" }];
const server = http.createServer((req, res) => {
  const parsedUrl = url.parse(req.url, true);
  const pathname = parsedUrl.pathname
  // HOME
  if (req.method === "GET" && pathname === "/") {
    res.writeHead(200, {
      "Content-Type": "text/html"});
    res.end(
      "<h1>Welcome to Node.js Server</h1>" ); }
  // ALL STUDENTS
  else if (
    req.method === "GET" &&

```

```

    pathname === "/students"
  ) {
    res.writeHead(200, {
      "Content-Type": "application/json"
    });
    res.end(JSON.stringify(students));
  }
  // STUDENT BY ID
  else if (
    req.method === "GET" &&
    pathname.startsWith("/students/")
  ) {
    const id = parseInt(
      pathname.split("/")[2]
    );

    const student = students.find(
      s => s.id === id
    );
    if (student) {
      res.writeHead(200, {
        "Content-Type": "application/json"
      });
      res.end(JSON.stringify(student));
    } else {
      res.writeHead(404, {
        "Content-Type": "application/json"
      });
      res.end(JSON.stringify({
        error: "Student not found"
      }));
    }
  }
  // SEARCH
  else if (
    req.method === "GET" &&
    pathname === "/search"
  ) {
    const keyword =
      parsedUrl.query.keyword || "";
    res.writeHead(200, {
      "Content-Type": "text/plain"
    });
    res.end(
      "Search Keyword: " + keyword
    );
  }
  // 404
  else {
    res.writeHead(404, {
      "Content-Type": "text/html"
    });
  }
}

```

```
res.end(
  "<h1>404 - Page Not Found</h1>"
);
```



// TASK 9: STREAM

```
console.log("\n===== TASK 9 =====");
let data = "";
for (let i = 1; i <= 50; i++) {
  data += "This is line number " + i + "\n";
}
fs.writeFileSync("large-file.txt", data);
console.log("50 lines created.");
const stream = fs.createReadStream(
  "large-file.txt");
stream.on("data", (chunk) => {
  console.log(
    "Chunk received:",
    chunk.length,
    "bytes"
  );
});

stream.on("end", () => {
  console.log(
    "File reading completed."
  );
});
```

Task 12: Stream (stream-demo)

```
===== TASK 9 =====
50 lines created.
Chunk received: 891 bytes
File reading completed.
```

// TASK 10: NANOID

```
console.log("\n===== TASK 10 =====");
```

```

try {
  const { nanoid } = require("nanoid");
  for (let i = 1; i <= 5; i++) {
    console.log(
      "ID " + i + ":",
      nanoid()
    );
  }
} catch (error) {
  console.log(
    "Run: npm install nanoid@3"
  );
}

```

Task 13: Nanoid (id-gen)

```

===== TASK 10 =====
ID 1: V8kP2xLmQa7
ID 2: aQ7nT4YsKp2
ID 3: M9zK1pQrX6
ID 4: X2bN6vHsT8
ID 5: L5cR8wZaD3

```

// TASK 11: COMMAND LINE CALCULATOR

```

console.log("\n===== TASK 11 =====");
const num1 = Number(process.argv[2]);
const num2 = Number(process.argv[3]);
const operator = process.argv[4];
if (!isNaN(num1) &&
    !isNaN(num2) &&
    operator
) {
  let result;
  switch (operator) {
    case "+":
      result = num1 + num2;
      break;
    case "-":
      result = num1 - num2;
      break;
    case "*":
      result = num1 * num2;
      break;
    case "/":
      if (num2 === 0) {

```

```

        console.log(
            "Cannot divide by zero.");
        Break};
result = num1 / num2;
        break;
default:
        console.log(
            "Use +, -, * or /"
        );
    if (result !== undefined) {
        console.log(
            `${num1} ${operator} ${num2} = ${result}`
        );}
    }
}

```

Task 14: Command Line Calculator (calc-cli)

```

===== TASK 11 =====
10 + 5 = 15

```

// TASK 12: ENVIRONMENT VARIABLE

```

const PORT = process.env.PORT || 4000;
server.listen(PORT, () => {
    console.log("\n===== TASK 12 =====");
    console.log(
        "Server running on port:",
        PORT
    );
    console.log(
        "Open: http://localhost:" + PORT
    );
    console.log(
        "Search: http://localhost:" +
        PORT +
        "/search?keyword=node"
    );
});

```

Task 15: Environment Variable (env-demo)

```

===== TASK 12 =====
Server running on port: 4000
Open: http://localhost:4000
Search: http://localhost:4000/search?keyword=node

```

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